

Biaglow, Andrew Dr

From: Nagelli, Enoch Dr
Sent: Monday, April 6, 2026 8:15 AM
To: Paul@the-dietrichs.com; armstm@udel.edu; liberatorem@trine.edu; kmschultz@purdue.edu; Admin Assistant; hair17@comcast.net; kisondra@gmail.com; kevin.a.shipe@gmail.com; michaeldeforest@fortna.com; m.theising@gmail.com; mdeforest@controltouch.com; kmachultz@purdue.edu; Paul@the-dietrichs.com; m.theising@gmail.com; gautham.krishnamoort@und.edu
Cc: Rogers, Nijel MAJ; Biaglow, Andrew Dr; Yuk, Simuck F Dr; Costain, Kristen Mrs; Jaffett, Victor A LTC
Subject: USMA ChemE Advisory Board Data Pack Readahead
Attachments: Advisory Board Meeting_Detailed.pdf; AY25 Program Assessment Data - 16 July 2025.pdf; Advisory Board Slides.pptx; Advisory Board Slides.pdf

Dear Board Members,

Attached is a detailed Advisory Board meeting schedule, AY25 Program Assessment Data, and a slide deck with key assessment data for your reference as a readahead for our meeting this week. We look forward to seeing you on Thursday and thank you again for your service to the Program and the Academy.

Safe Travels and please feel free to contact me email or call/text at 630-452-3653 if you have any questions.

Looking forward to our discussions!

Sincerely,

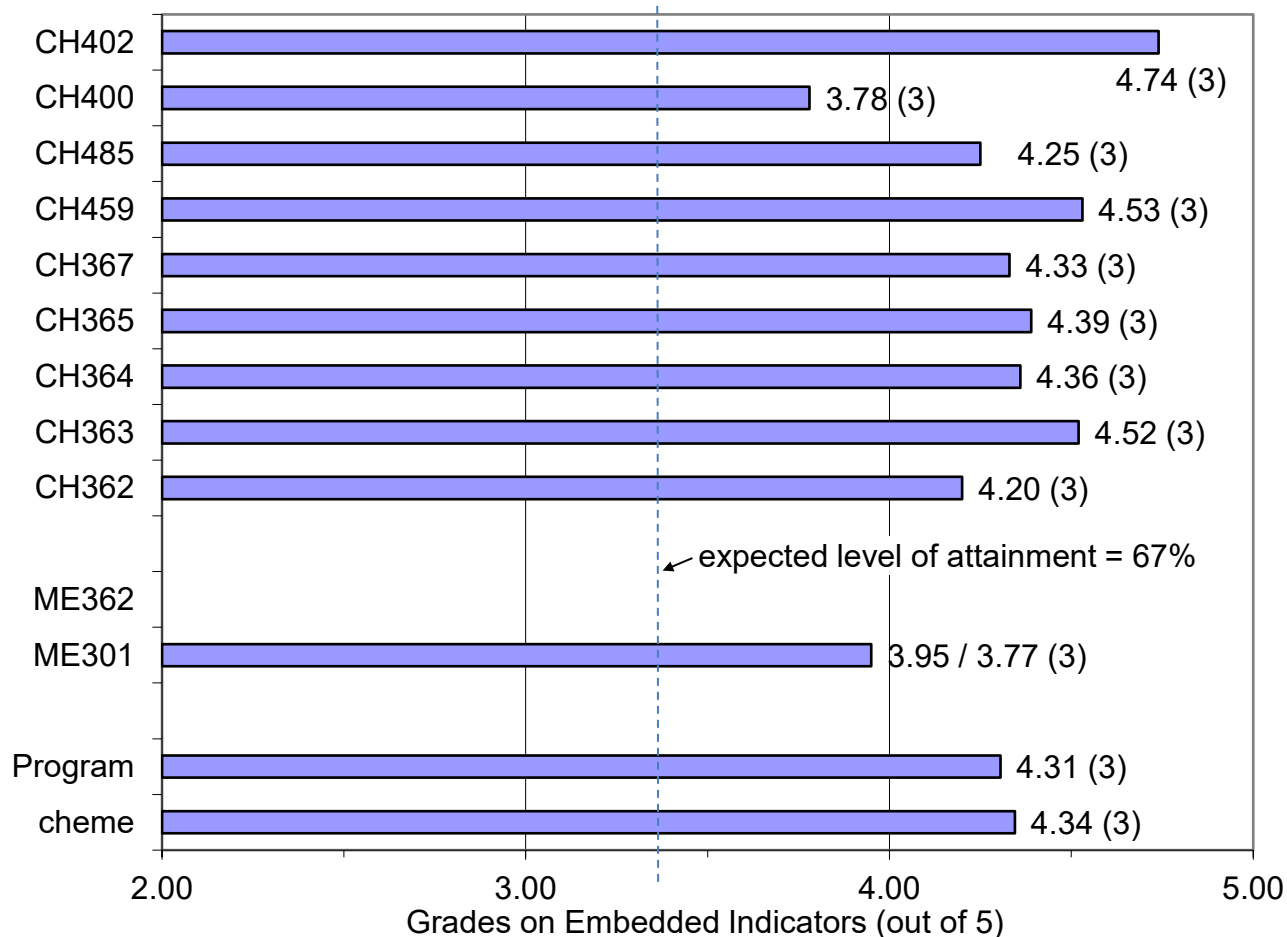
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Example Data: Coursework Embedded Indicators

Student Outcome 1 – Example Data

Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.



Values in parentheses are coverage ratings from Table 5-4 in the 2020 Self-Study and explained on page 5-7.

Parentheses ratings:.

3: Unique embedded indicator with clear rubric or cut scale.

2: Outcome was graded but grades are convoluted, or part of the outcome is not covered.

1: Correlation to outcome but no assessment

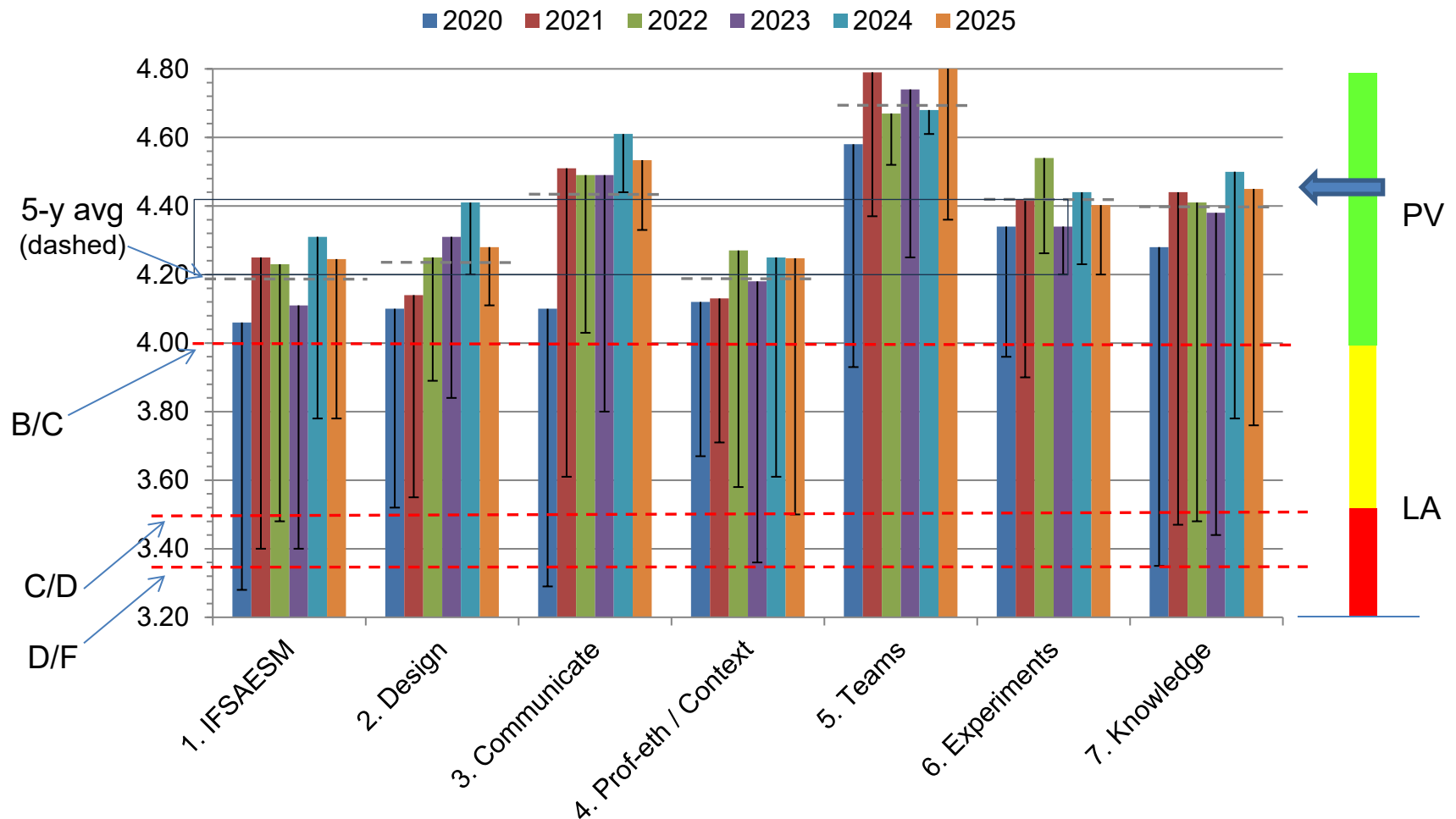
0: No coverage or correlation

Data shown here is for Class of 2025
Similar data is collected for all 7 ABET student outcomes
Summary of all data is shown on next slide



Performance on Embedded Indicators

Program Averages AY2020 to AY2025



All indicators are in the green and trending upward.

Error bars are minimum scores from course assessments.

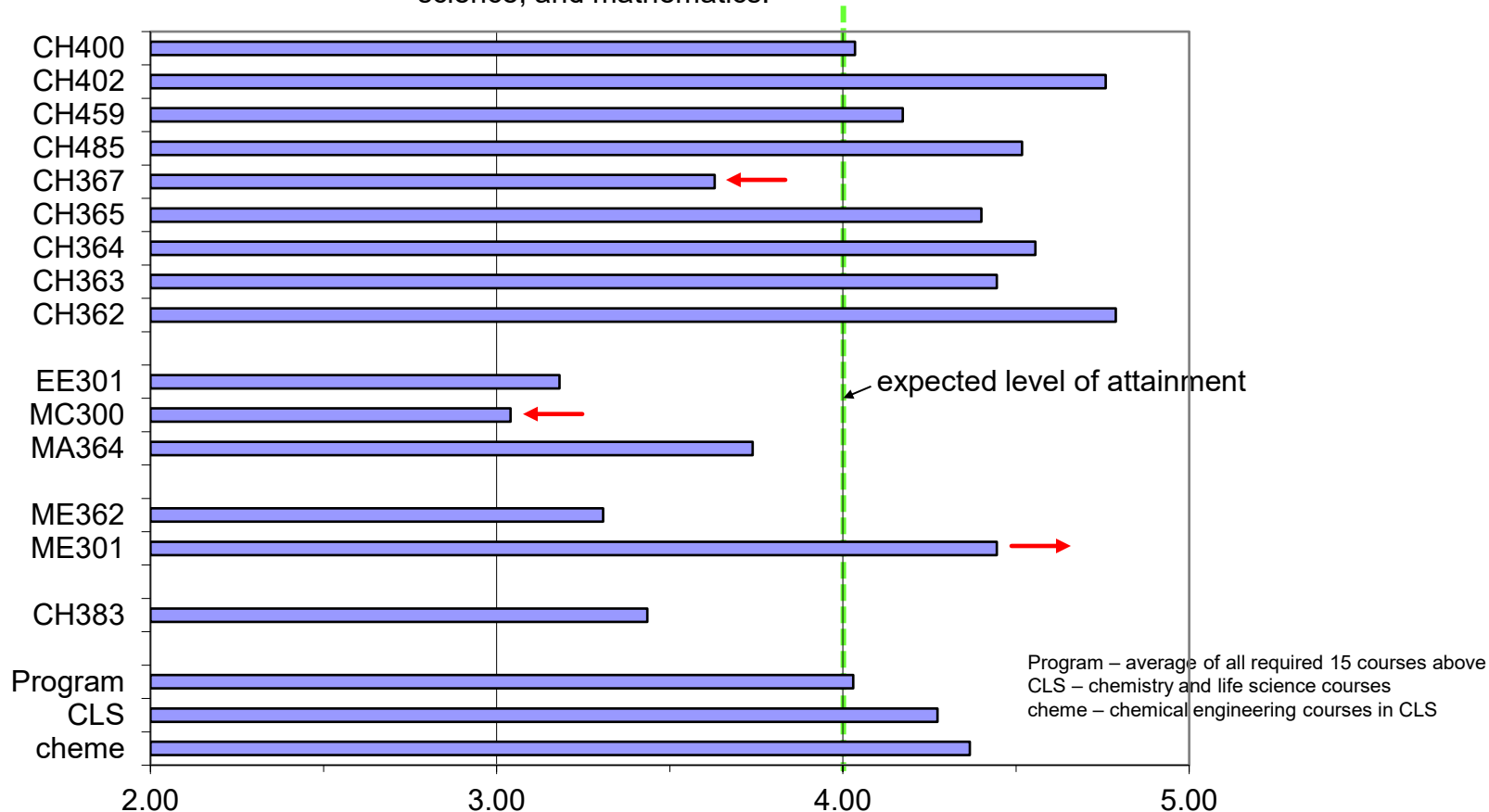
Expected levels of attainment are the 5-year averages



Example Data: End-of-Semester Surveys

Student Outcome 1

This course has improved my ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

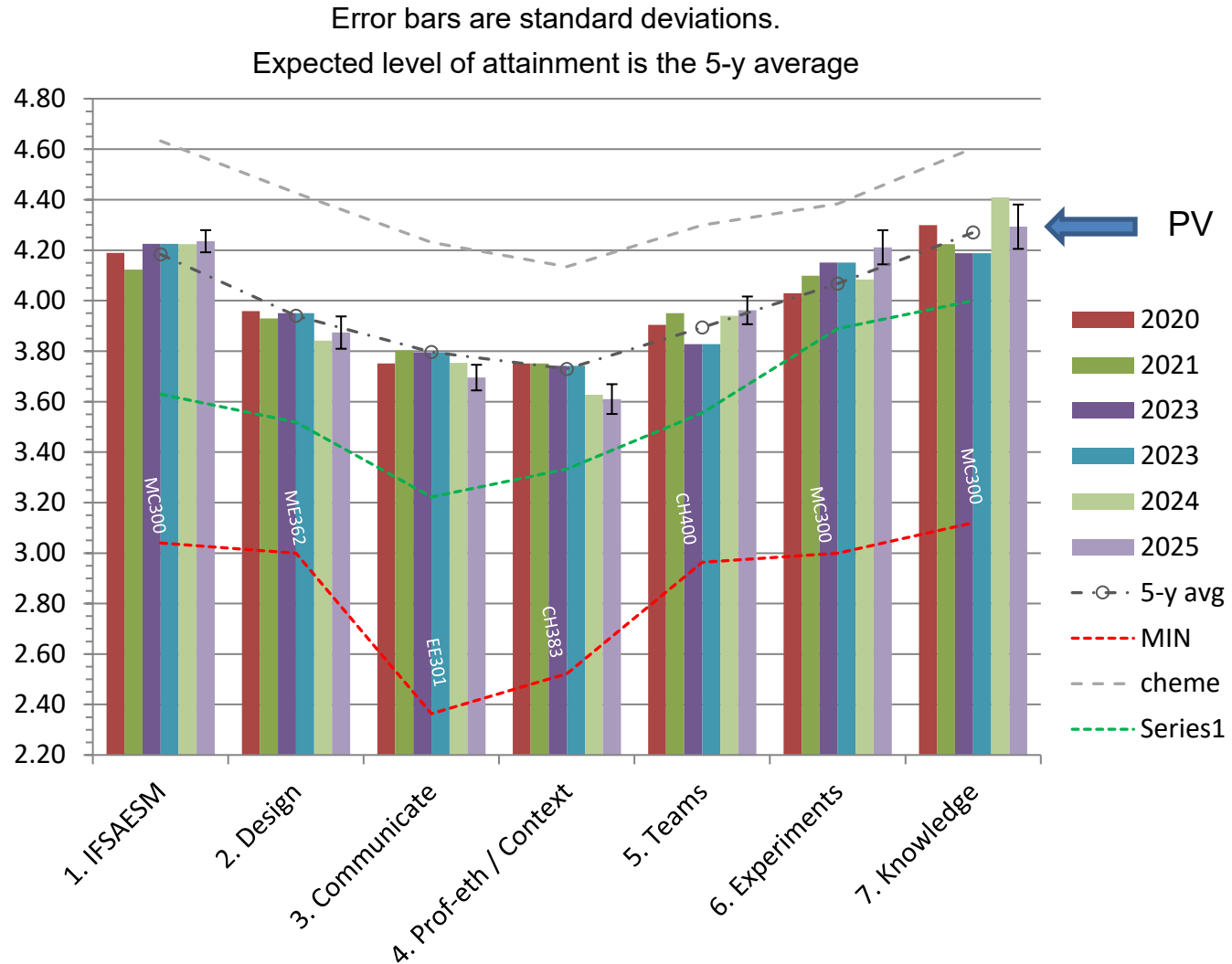


Data shown here is for Class of 2024
Similar data is collected for ABET student outcomes 1-7
Summary of all data is shown on next slide



End-of-Semester Surveys

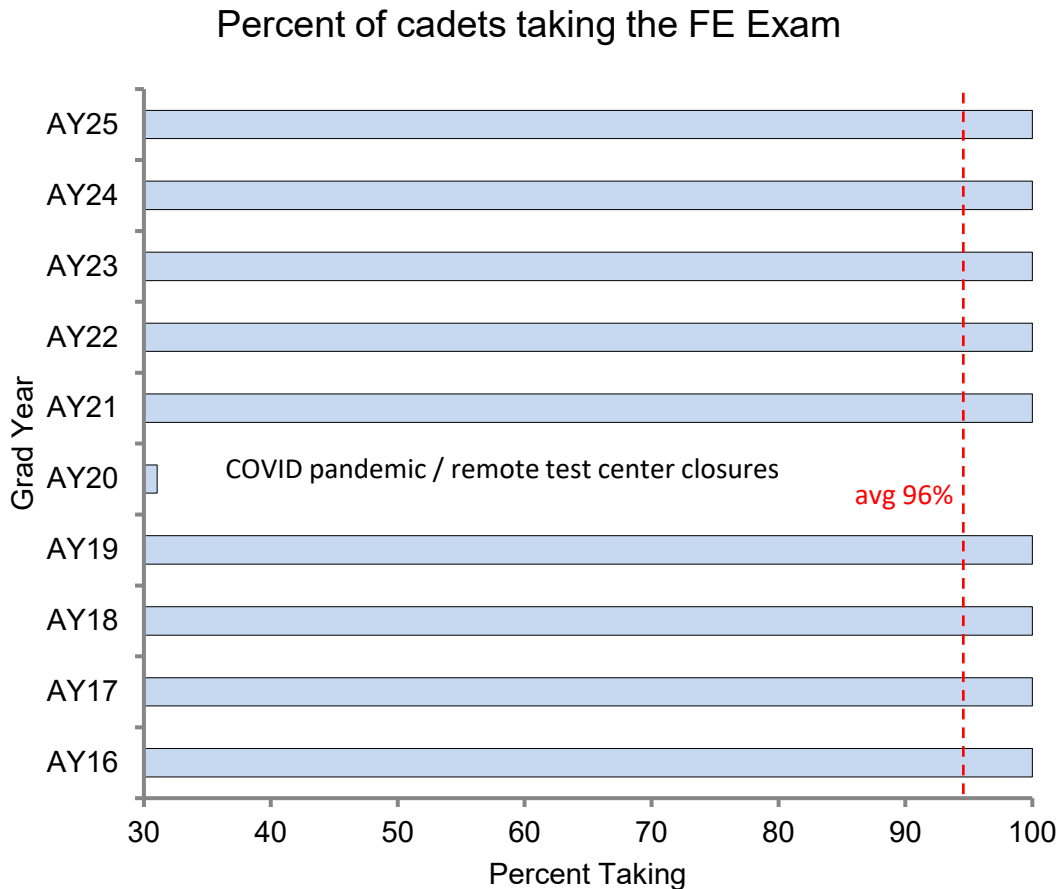
Program Averages from AY20-25





Fundamentals of Engineering Exam

Student Outcome 7: Acquire and apply new knowledge as needed, using appropriate learning strategies

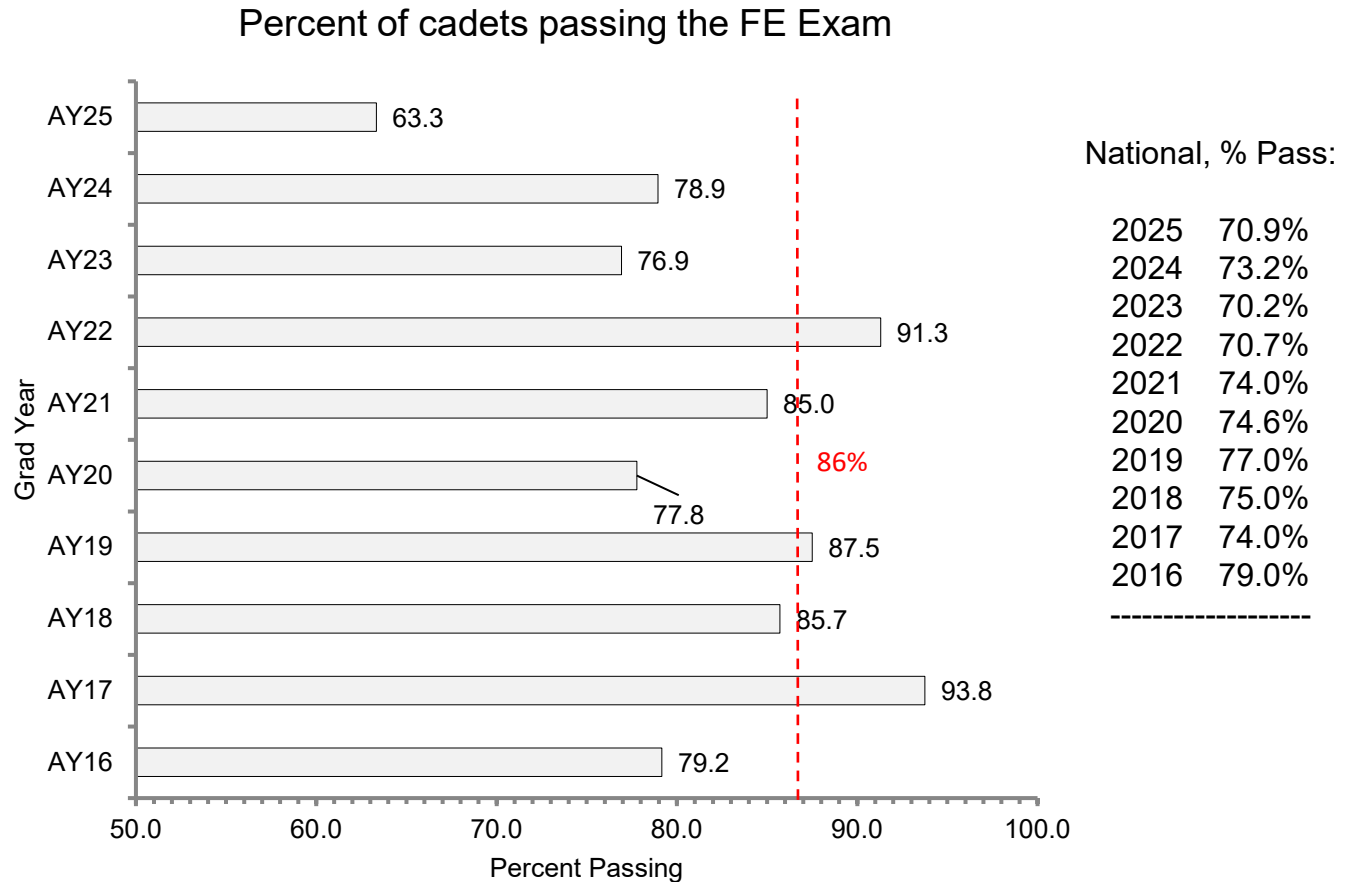


The average take rate was 86% prior to introduction of CH400 in AY2010.



Fundamentals of Engineering Exam

Student Outcome 1: Identify, formulate, and solve problems by applying principles of engineering, science, and mathematics.



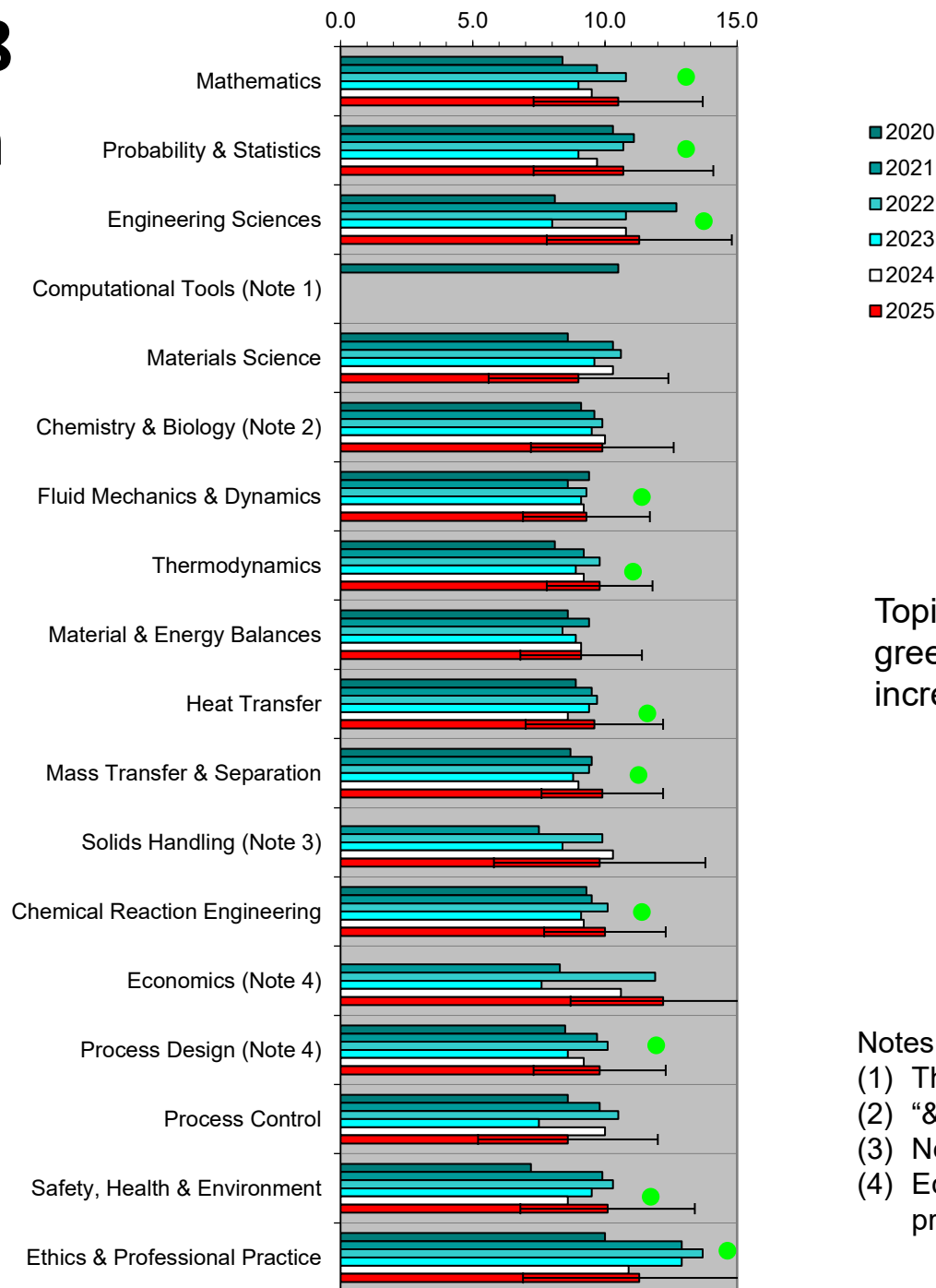
Outcome 8 Evaluation

FEE Results by Topic

AY20 to AY25

The error bars are
the individual
standard deviations
for the AY25 data.

The average
standard deviation
over all data is 3.0.



Topics labeled with
green circles (●)
increased from last AY.

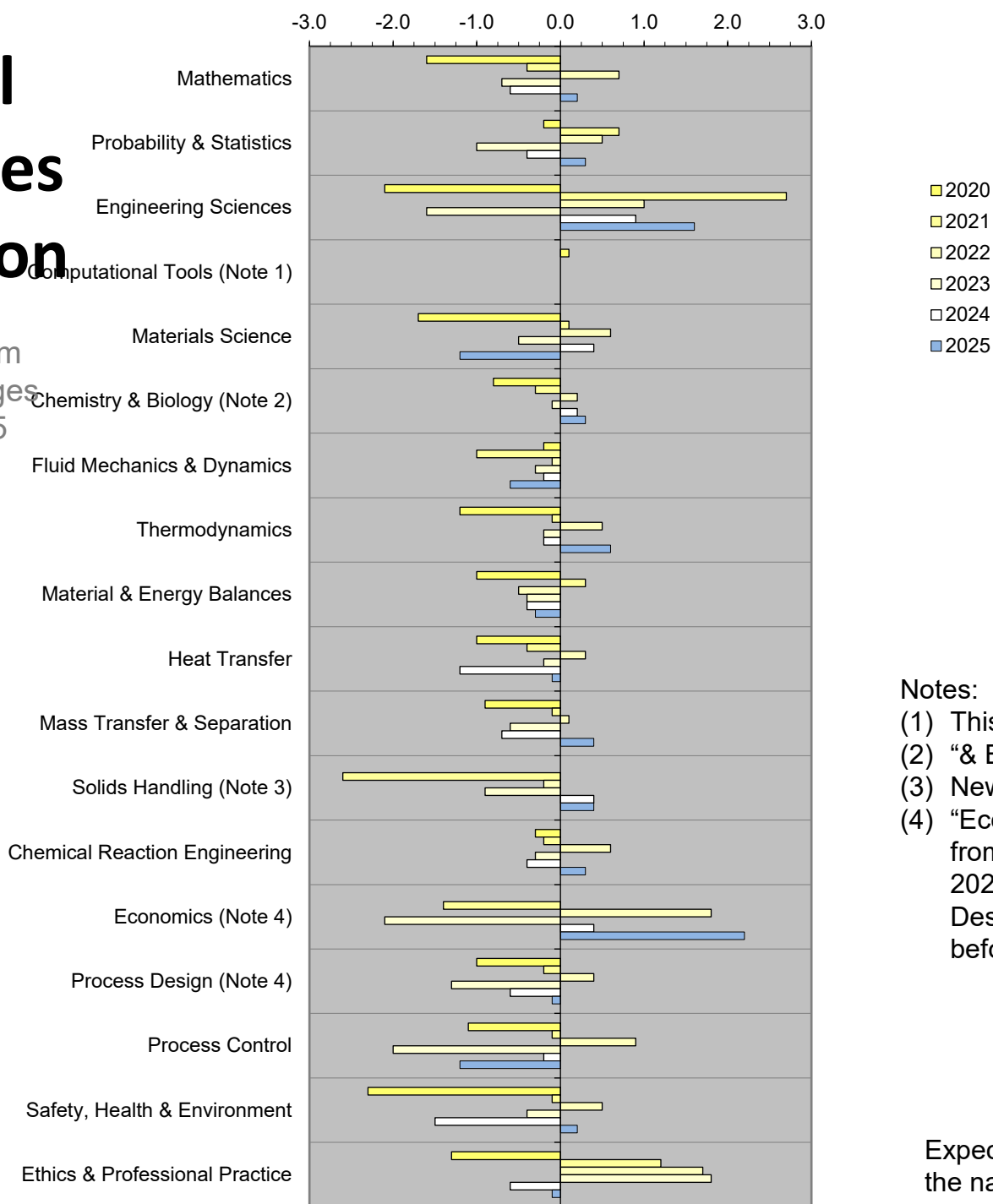
Notes:

- (1) This topic dropped in 2021.
- (2) "& Biology" added in 2021
- (3) New exam spec in 2021
- (4) Economics separated from process design in 2021.



Topical Outcomes Evaluation

Deviations from
National Averages
AY20 to AY25



Notes:

- (1) This topic dropped in 2021.
- (2) "& Biology" added in 2021
- (3) New exam spec in 2021
- (4) "Economics" separated from process design in 2021. Topic was Proc Design & Econ 2020 and before

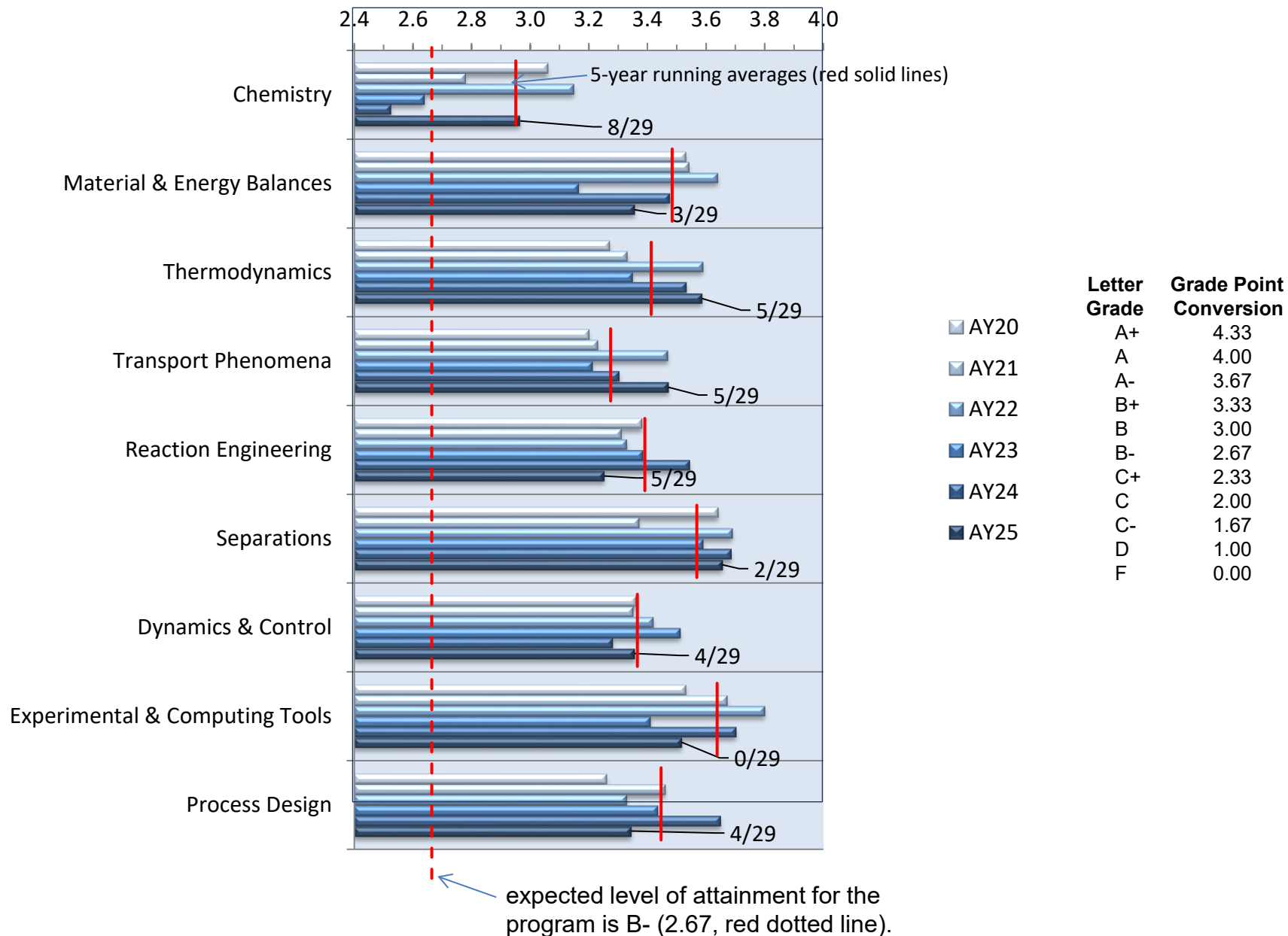
Expected level of attainment is the national average (0.0 line)



Topical Outcomes Evaluation

Student Outcome 8: Understanding of the Chemical Engineering Curriculum

Average GPA from Transcripts, AY2020 to AY2025



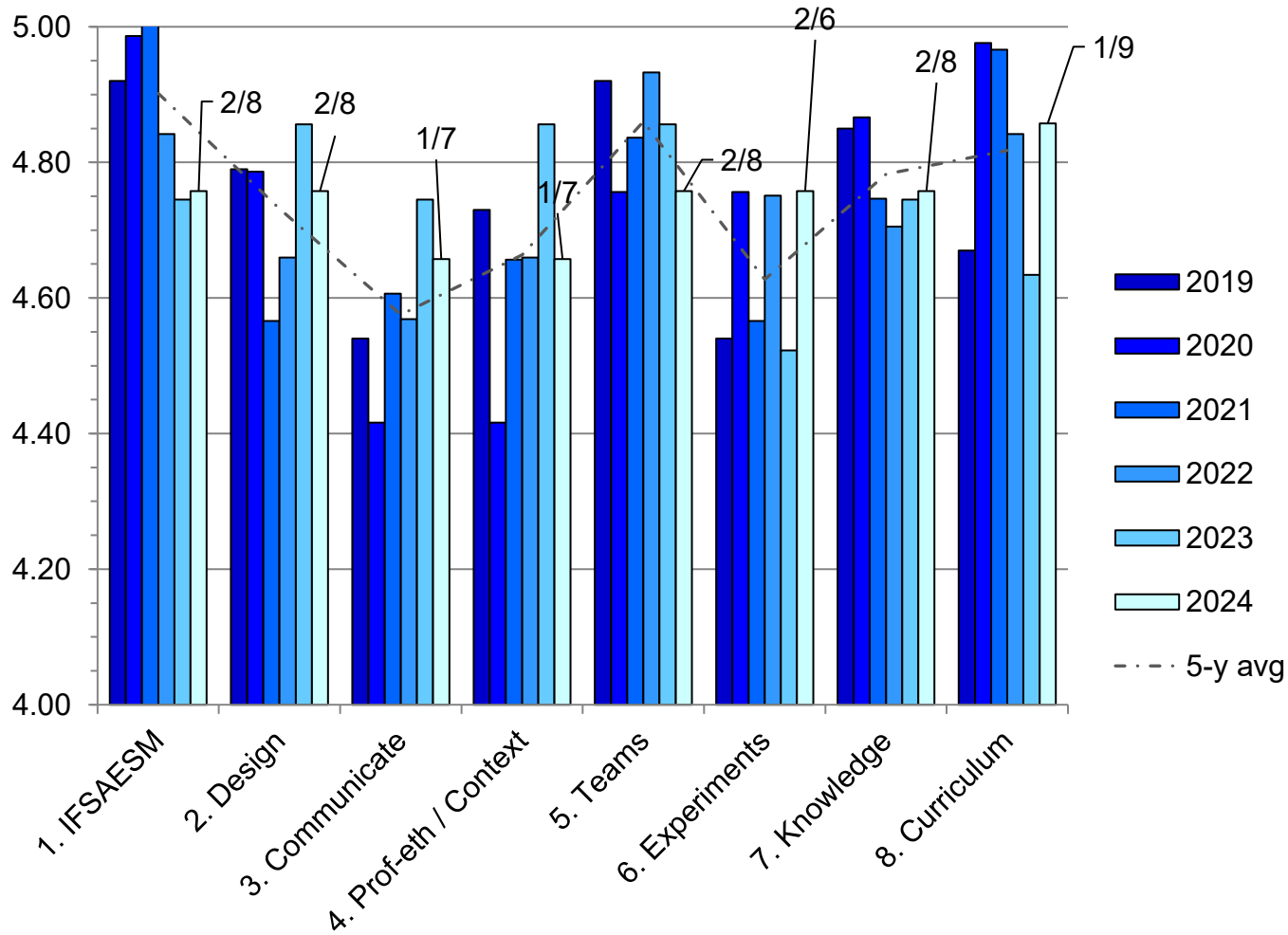


Advisory Board Student Outcomes Surveys

Student Outcomes 1-8

Program Averages from AY19-24

Data labels are response frequencies for 4 or 5 (# of 4s / # of 5s) on the 1-5 Survey Likert Scale
Standard deviations range from .00 to .52



AY25 data will be available after the next advisory board meeting in Spring 2026.

PROGRAM ASSESSMENT DATA AY2025

UNITED STATES MILITARY ACADEMY
DEPARTMENT OF CHEMISTRY AND LIFE SCIENCE
CHEMICAL ENGINEERING PROGRAM
July 16, 2025

<u>Student Outcome</u>	<u>Page</u>
1.....	<u>1</u>
2.....	<u>3</u>
3.....	<u>5</u>
4.....	<u>7</u>
5.....	<u>9</u>
6.....	<u>12</u>
7.....	<u>14</u>
8.....	<u>19</u>

Level of Achievement of Student Outcome 1:

On completion of the chemical engineering program, our graduates will be able to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. Fundamentals of Engineering Examination, once/yr.
3. End-of-Semester Student Surveys, once/semester.
4. Chemical Engineering Program Exit Survey, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

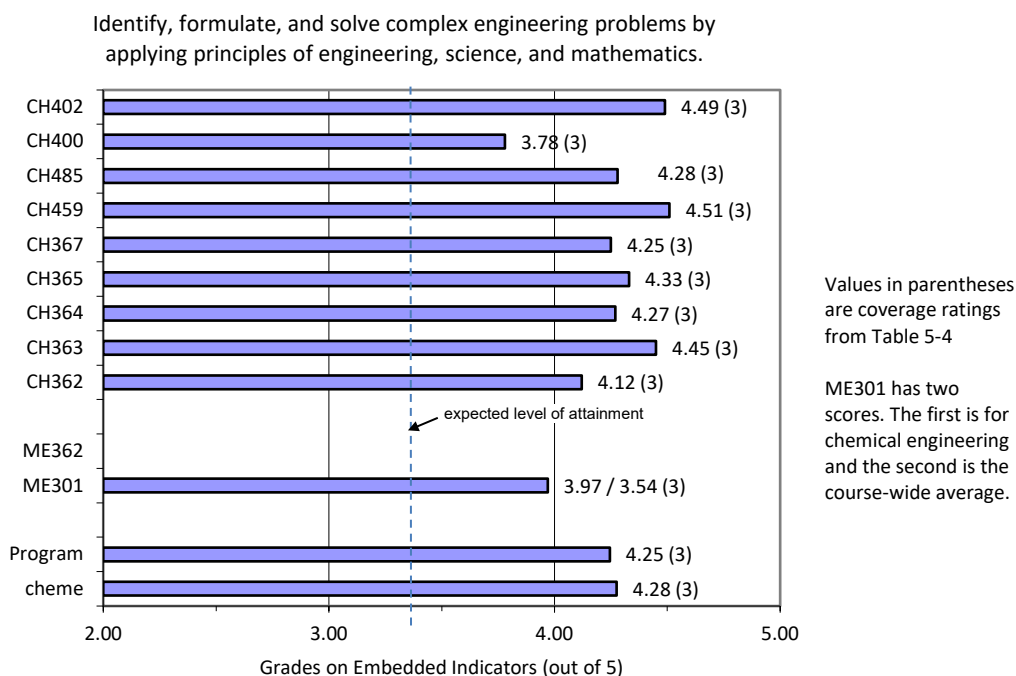
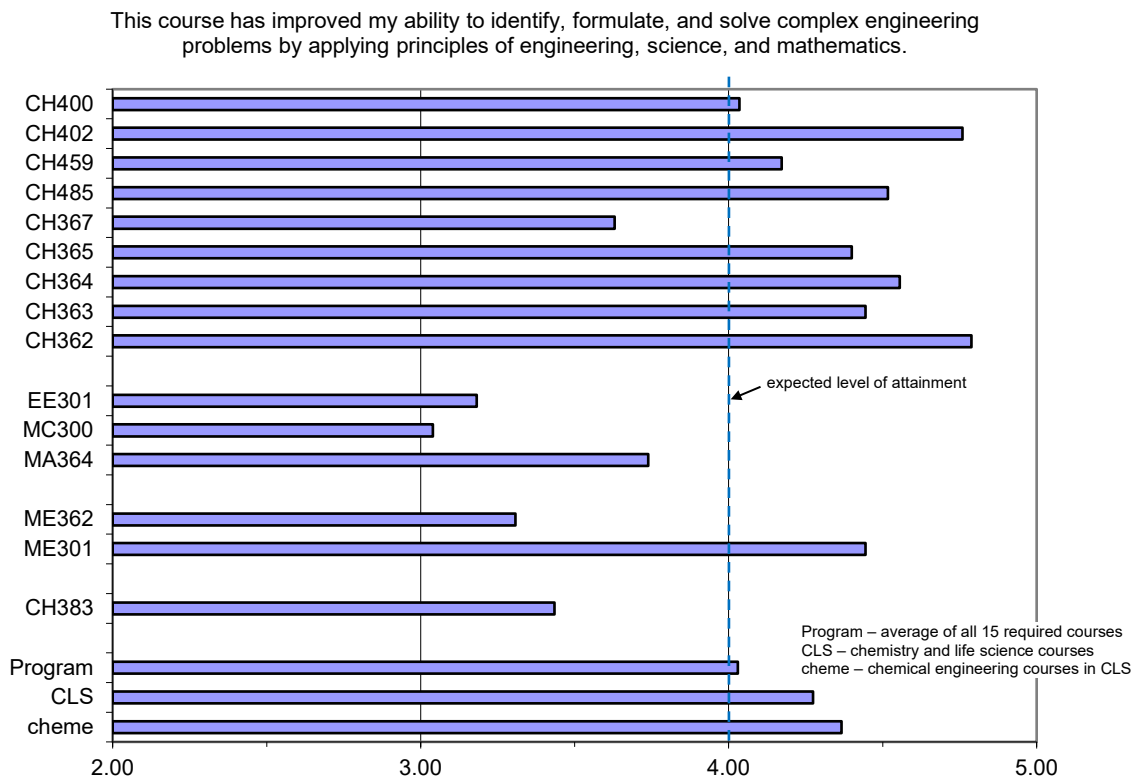


Figure 1-1. Coursework Embedded Indicator results for Student Outcome 1.

2. Fundamentals of Engineering Examination (FEE). According to the 2025 report from NCEES, 18 out of 30 cadets in the Class of 2025 took and passed the FEE. After one successful re-take, our program is at 19 out of 30, or 63.3%. The national average passing in 2025 was 70.9%, and this is our expected level of attainment. In the previous five years, the pass rates were 78.9% in 2024, 76.9% in 2023, 91.3% in 2022, 85.0% in 2021, and 77.8% in 2020. Our running average over those previous five years is $82.0\% \pm 6.1\%$ (versus $72.6\% \pm 2.0\%$ for the national average). The national average is our expected level of attainment.

3. End of Semester Student Surveys

Figure 1-2. End-of-Semester Student Survey responses for Student Outcome 1.



4. Chemical Engineering Program Exit Survey. This survey is issued to Firsties at the end of their last semester. Cadets were asked whether they agree with the statement “The program has prepared me to Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.” 29 out of 29 cadets completed the survey. Of these, 29 out of 29 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 21/29 replying that they strongly agreed (score = 5/5) and 8/29 that they agreed (score = 4/5). The mean is $4.72/5.00 \pm 0.46$ for the 29 cadets participating in the survey. The expected level of attainment on this survey is 4.00/5.00.

Level of Achievement of Student Outcome 2:

On completion of the chemical engineering program, our graduates will be able to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End of Semester Student Surveys, once/semester.
3. Course Grades in CH402 Chemical Engineering Process Design, once/yr.
4. Chemical Engineering Program Exit Survey, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

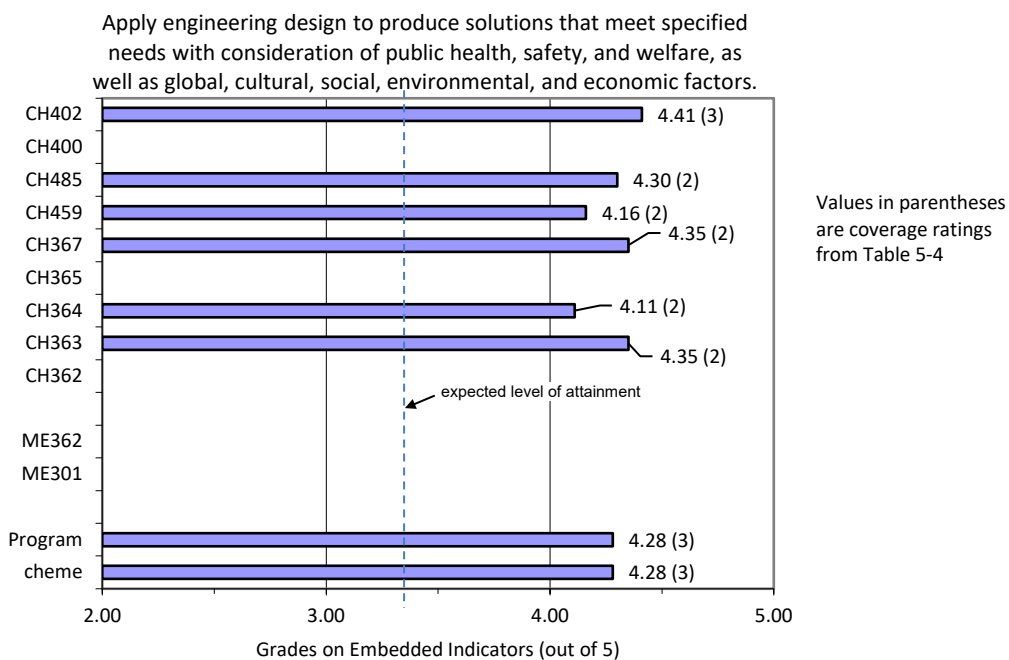


Figure 2-1. Coursework Embedded Indicator results for Student Outcome 2.

2. End of Semester Student Surveys

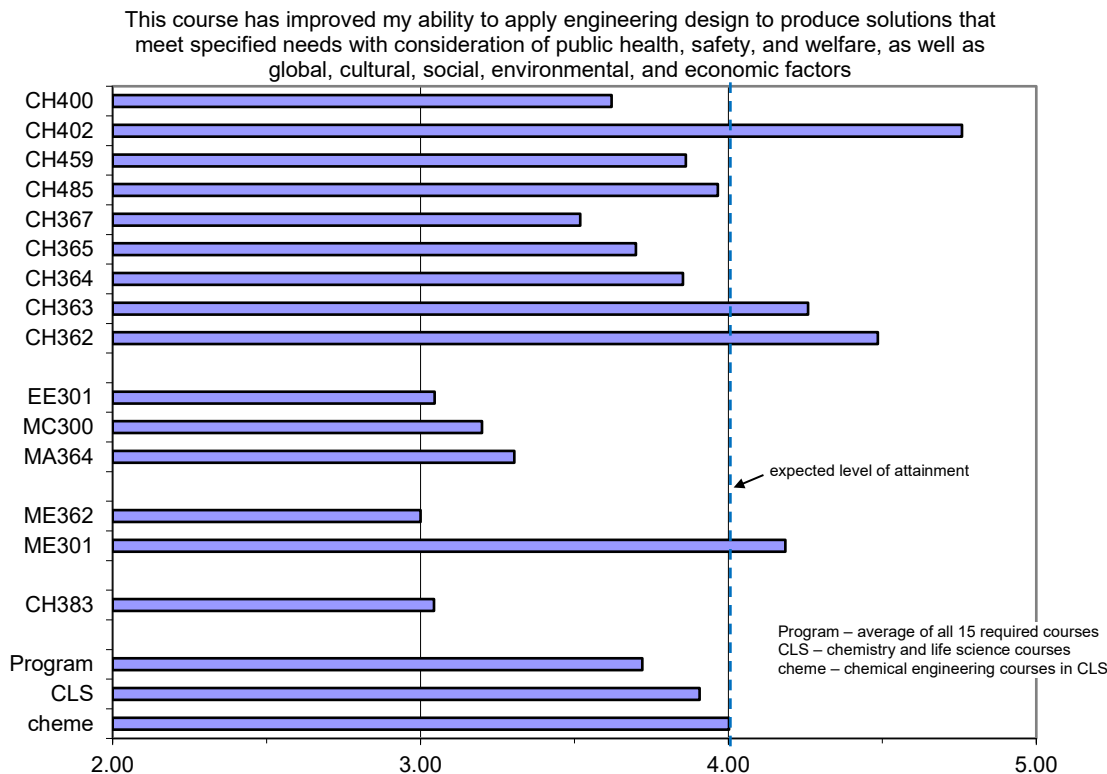


Figure 2-2. End-of-Semester Student Survey responses for Student Outcome 2.

- The average course grade in CH402 Chemical Engineering Process Design was 3.34 ± 0.70 ($n=29$) in AY25, compared to 3.65 ± 0.46 ($n=19$) in AY24, 3.44 ± 0.55 ($n=13$) in AY23, 3.33 ± 0.50 ($n=23$) in AY22, 3.46 ± 0.68 ($n=20$) in AY21, and 3.26 ± 0.70 ($n=29$) in AY20. The 5-year running average for the previous five years is 3.43, and this is our expected level of attainment. This year's score was *slightly below* the 5-year running average.
- Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to Firsties at the end of their last semester. They were asked whether or not they agree with the statement "The program has prepared me to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors." 29 out of 29 cadets completed the survey. Of these, 29 out of 29 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 15/29 replying that they strongly agreed (score = 5/5) and 14/29 that they agreed (score = 4/5). The mean is $4.52/5.00 \pm 0.51$ for the 29 cadets. The expected level of attainment on this survey is 4.00/5.00.

Level of Achievement of Student Outcome 3:

On completion of the chemical engineering program, our graduates will be able to communicate effectively with a range of audiences.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End of Semester Student Surveys, once/semester.
3. Course Grades in CH459 Unit Operations Laboratory, once/yr.
4. Chemical Engineering Program Exit Survey, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

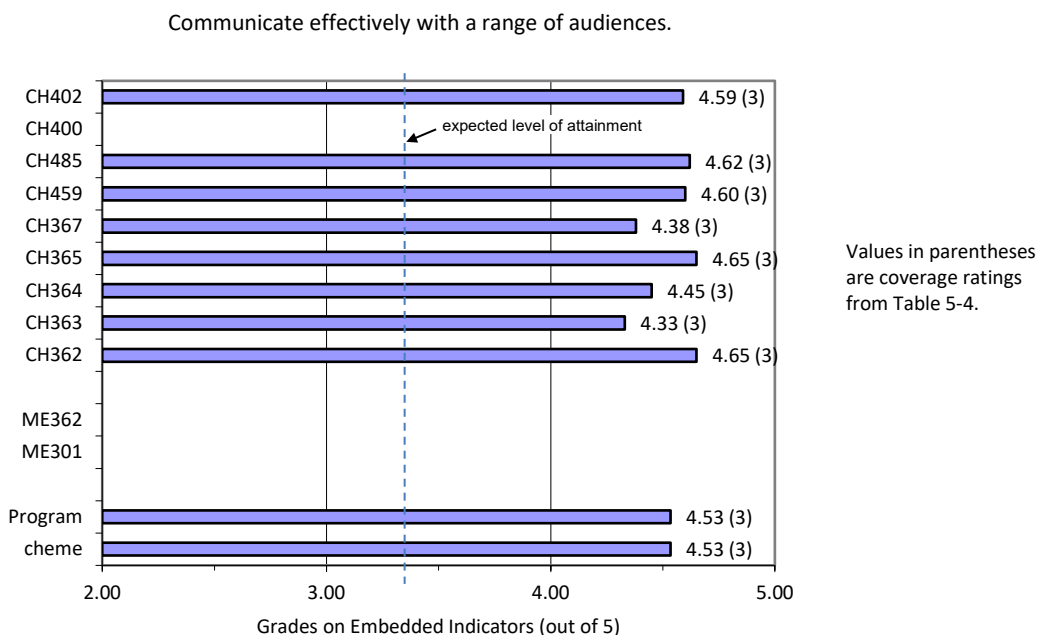


Figure 3-1. Coursework Embedded Indicator results for Student Outcome 3.

2. End of Semester Student Surveys

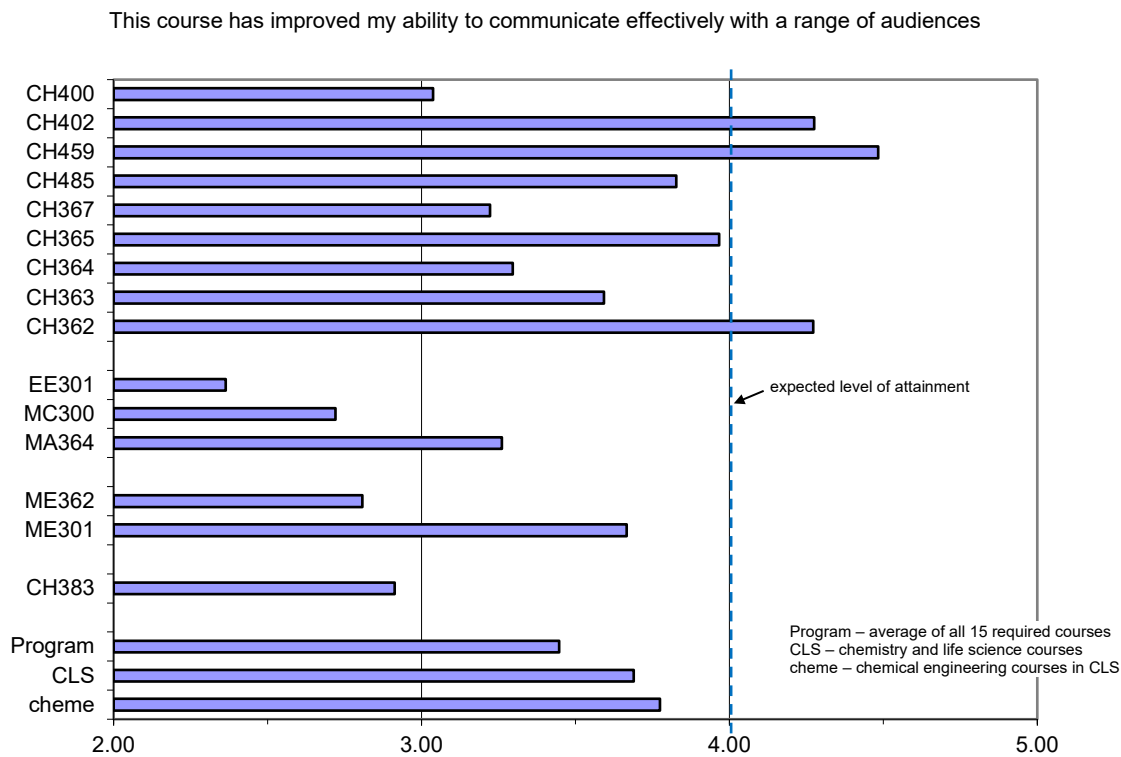


Figure 3-2. End-of-Semester Student Survey responses for Student Outcome 3.

3. The average course grade in CH459 Chemical Engineering Laboratory was 3.52 ± 0.59 ($n=29$) in AY25, compared to 3.70 ± 0.46 ($n=19$) in AY24, 3.41 ± 0.43 ($n=13$) in AY23, 3.80 ± 0.42 ($n=23$) in AY22, 3.67 ± 0.53 ($n=18$) in AY21, and 3.53 ± 0.48 ($n=29$) in AY20. The 5-year running average is 3.62, and this is our expected level of attainment. This year's score was *slightly below* the 5-year running average.
4. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to Firsties at the end of their last semester. They were asked whether they agree with the statement "The program has prepared me to communicate effectively with a range of audiences." 29 out of 29 cadets completed the survey. Of these, 27 out of 29 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 13/29 replying that they strongly agreed (score = 5/5), 11/29 that they agreed (score = 4/5), and 2/29 that they were neutral. The mean is $4.38/5.00 \pm 0.62$ for the 29 cadets. The expected level of attainment on this survey is 4.00/5.00.

Level of Achievement of Student Outcome 4:

On completion of the chemical engineering program, our graduates will be able to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. Fundamentals of Engineering Examination Performance Index, once/yr.
3. End of Semester Student Surveys, once/semester.
4. Chemical Engineering Program Exit Survey, once/yr.
5. Completion of Cadet Character Education Program, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

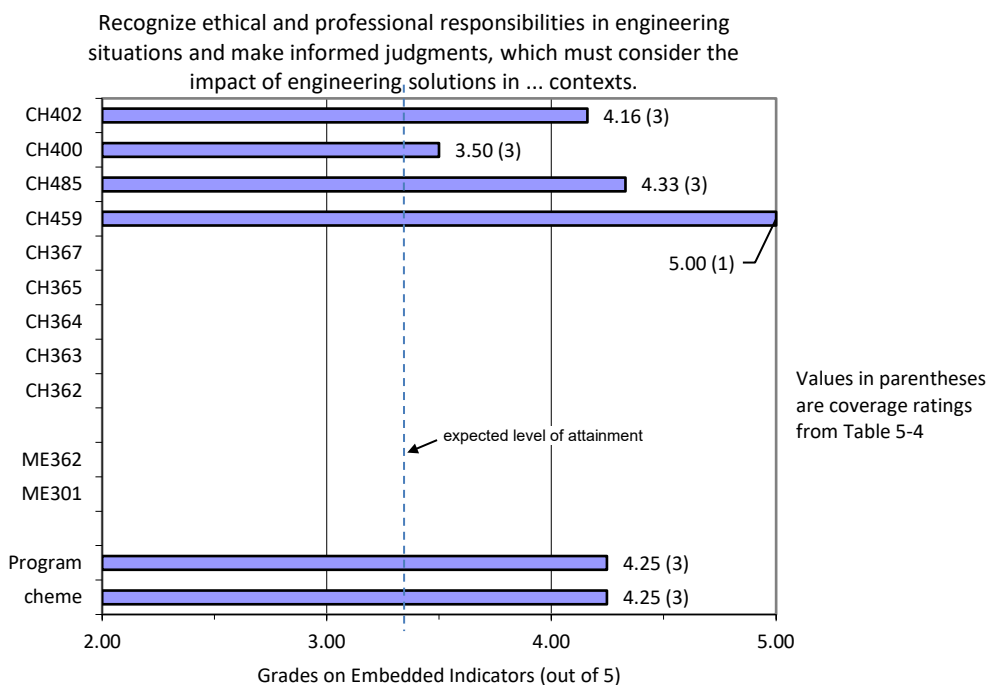


Figure 4-1. Coursework Embedded Indicator results for Student Outcome 4.

2. Fundamentals of Engineering Examination Performance, Self-Study Table 4-2.

Subject	Outcome	Questions	USMA	National (Expected level of attainment)
Ethics and Professional Practice	4	3	11.3	11.4 ± 4.3
Economics	4	4	12.2	10.0 ± 3.7

The national average is our expected level of attainment.

3. End of Semester Student Surveys

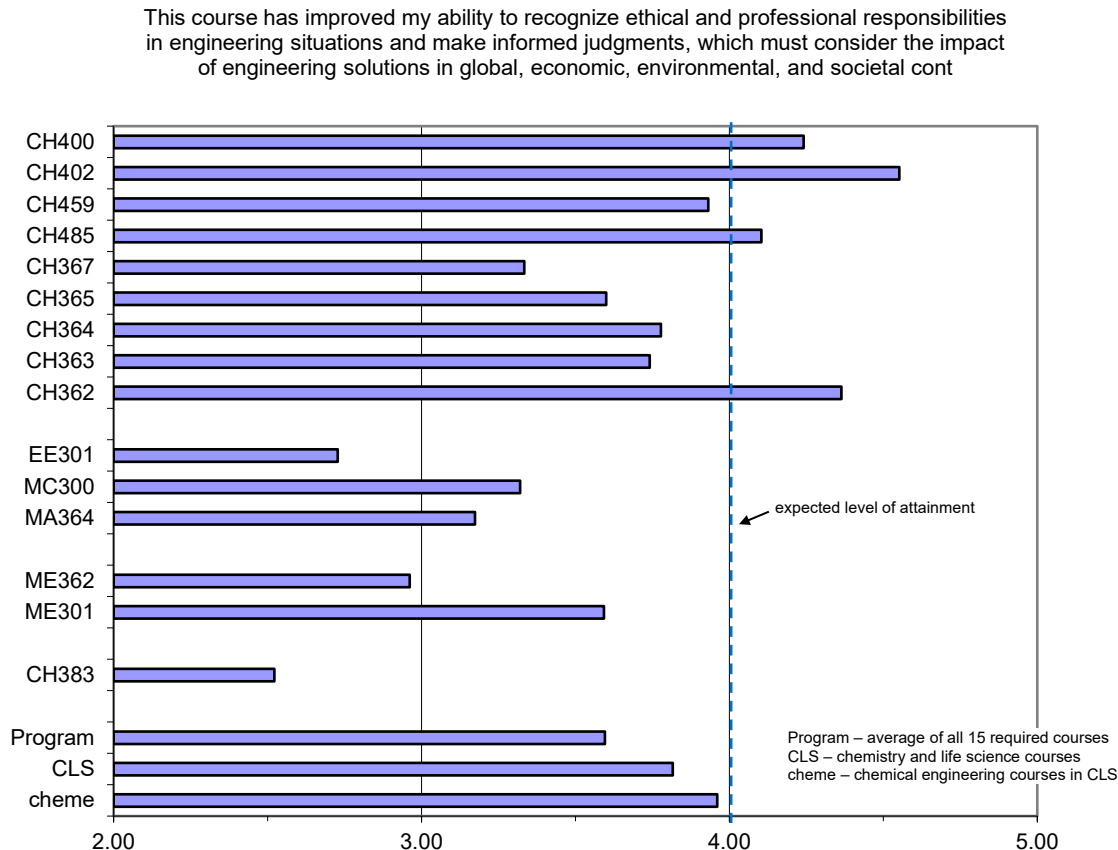


Figure 4-2. End-of-Semester Student Survey responses for Student Outcome 4.

4. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to Firsties at the end of their last semester. They were asked whether they agree with the statement “The program has prepared me to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.” 29 out of 29 cadets completed the survey. Of these, 27 out of 29 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 11/29 replying that they strongly agreed (score = 5/5), 16/29 that they agreed (score = 4/5), and 2/29 that they were neutral. The mean is $4.31/5.00 \pm 0.60$ for the 29 cadets. The expected level of attainment is 4.00/5.00.
5. Training in honor and ethics takes place in the Cadet Character Education Program (CCEP) during the academic year and summer terms. The program is overseen by the Commandant of Cadets through the Simon Center for the Professional Military Ethic. CCEP customizes instruction to each of the four year-groups of cadets, who interact with faculty volunteers who share their perspectives and experience in the Armed Forces, with industry, and at other civilian institutions. All 29 chemical engineering cadets successfully completed the 4-year CCEP program.

Level of Achievement of Student Outcome 5:

On completion of the chemical engineering program, our graduates will be able to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End of Semester Student Surveys, once/semester.
3. Chemical Engineering Program Exit Survey, once/yr.
4. Teamwork Skills Rubric, twice/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

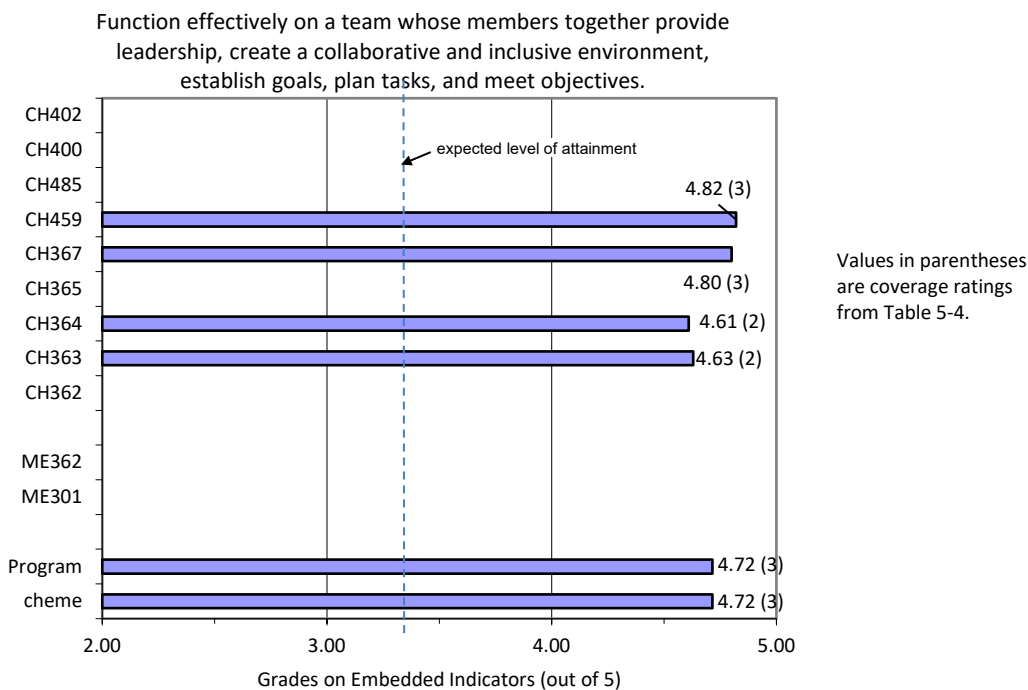


Figure 5-1. Coursework Embedded Indicator results for Student Outcome 5.

2. End of Semester Student Surveys

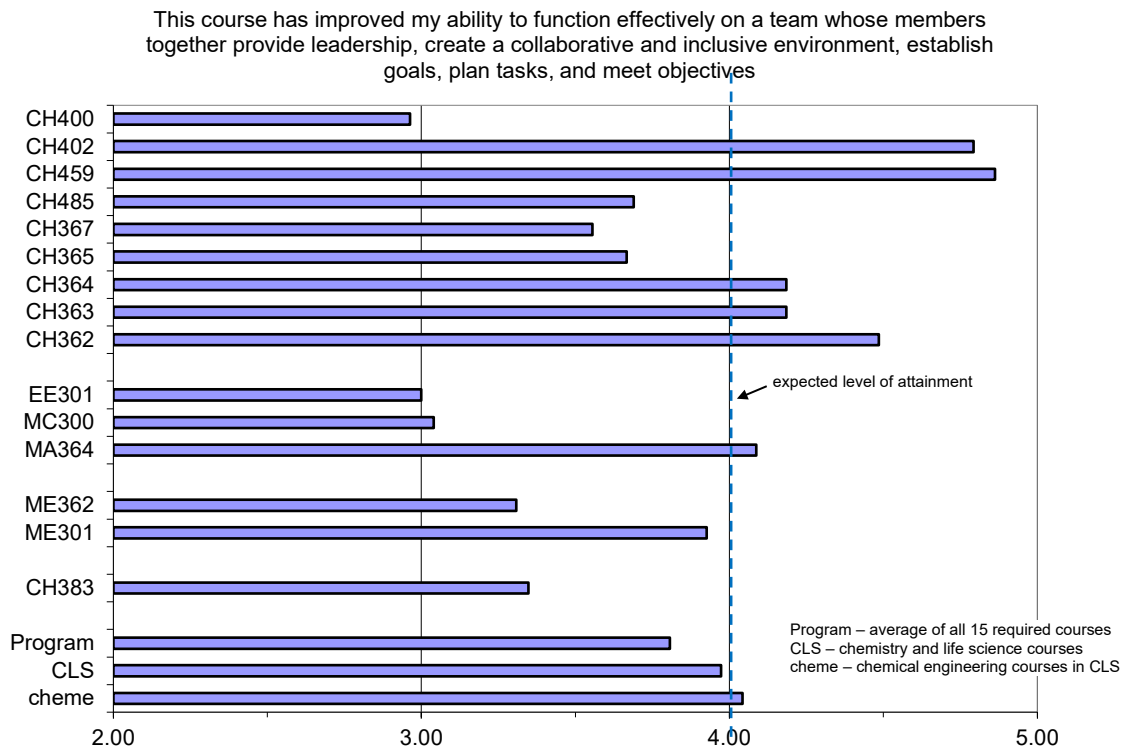


Figure 5-2. End-of-Semester Student Survey responses for Student Outcome 5.

3. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to the Firsties at the end of their last semester. They were asked whether or not they agree with the statement “The program has prepared me to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.” 29 out of 29 cadets completed the survey. Of these, 29 out of 29 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 24/29 replying that they strongly agreed (score = 5/5) and 5/29 that they agreed (score = 4/5). The mean is $4.83/5.00 \pm 0.38$ for the 29 cadets. The expected level of attainment on this survey is 4.00/5.00.
4. Teamwork Skills Rubric. The rubric is used to assess performance in four skills associated with the ability to function on effectively on a team, namely technical competence, communication, organization, and teamwork. The rubrics are completed by the cadets after each laboratory exercise in CH459. The results were averaged over each round-robin and reported below using the actual rubric for formatting. The superscript “a” next to the averages are for team assessment of the group leader (TAL) and “b” designates group leader assessment of the team (GLAT). The expected level of attainment is 4.0. The color shading indicates the relationship of the averages to the expected level of attainment. Red shading indicates when improvement is needed.

Your Name: Dr. Enoch Nagelli				Person Assessed: Cadets in CH459					
Your Position: CH459 CD				Major of Person Assessed: Chemical Engineering					
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5	N/A
Technical Competence	Some misunderstandings of the technical content.			Demonstrated knowledge of the technical content.			Exceptional knowledge of technical content.	4.7 ^a ±.19 ^c 4.8 ^b ±.03 ^c	
Communication	Lacked sensitivity and/or did not provide specific suggestions for improvement.			Effectively communicated important points.			Exceptional ability to explain important points. Very effectively communicated ideas for improvement.	4.7 ^a ±.22 ^c 4.7 ^b ±.11 ^c	
Organization	Was not prepared or did not give sufficient time to prepare.			Demonstrated effective organization during class.			Was exceptionally efficient, timely and responsive throughout the entire process.	4.7 ^a ±.16 ^c 4.8 ^b ±.16 ^c	
Teamwork	Demonstrated limited ability to see other perspectives or find common ground.			Worked collaboratively with team members to reach consensus.			Exceptional ability to help group find common ground or resolve conflict in order to ultimately reach consensus.	4.8 ^a ±.15 ^c 4.8 ^b ±.16 ^c	
Are the cadets capable of functioning on multidisciplinary teams? Yes	Comments: Each cadet was group leader twice. Footnote “a” designates the average of all Team Assessments of Leader (TAL) scores, while “b” designates average of all Group Leader Assessment (GLAT) scores. Footnote “c” designates standard deviations.						Assignment used for assessment: AY25, Round Robin 1		

Your Name: Dr. Enoch Nagelli				Person Assessed: Cadets in CH459					
Your Position: CH459 CD				Major of Person Assessed: Chemical Engineering					
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5	N/A
Technical Competence	Some misunderstandings of the technical content.			Demonstrated knowledge of the technical content.			Exceptional knowledge of technical content.	4.8 ^a ±.19 ^c 4.7 ^b ±.07 ^c	
Communication	Lacked sensitivity and/or did not provide specific suggestions for improvement.			Effectively communicated important points.			Exceptional ability to explain important points. Very effectively communicated ideas for improvement.	4.8 ^a ±.18 ^c 4.8 ^b ±.07 ^c	
Organization	Was not prepared or did not give sufficient time to prepare.			Demonstrated effective organization during class.			Was exceptionally efficient, timely and responsive throughout the entire process.	4.7 ^a ±.13 ^c 4.7 ^b ±.20 ^c	
Teamwork	Demonstrated limited ability to see other perspectives or find common ground.			Worked collaboratively with team members to reach consensus.			Exceptional ability to help group find common ground or resolve conflict in order to ultimately reach consensus.	4.8 ^a ±.16 ^c 4.8 ^b ±.08 ^c	
Are the cadets capable of functioning on multidisciplinary teams? Yes	Comments: Each cadet was group leader twice. Footnote “a” designates the average of all Team Assessments of Leader (TAL) scores, while “b” designates average of all Group Leader Assessment (GLAT) scores. Footnote “c” designates standard deviations.						Assignment used for assessment: AY25, Round Robin 2		

Level of Achievement of Student Outcome 6:

On completion of the chemical engineering program, our graduates will be able to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End-of-Semester Student Surveys, once/semester.
3. Chemical Engineering Program Exit Survey, once/yr.
4. Course Grades in CH459 Unit Operations Laboratory, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

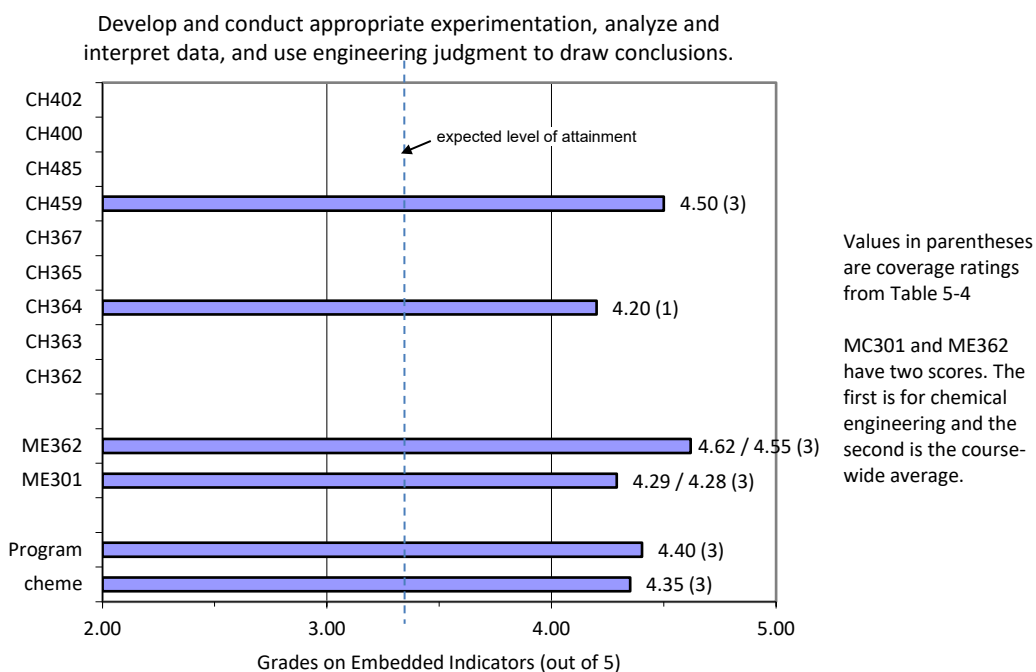


Figure 6-1. Coursework Embedded Indicator results for Student Outcome 6.

2. End of Semester Student Surveys

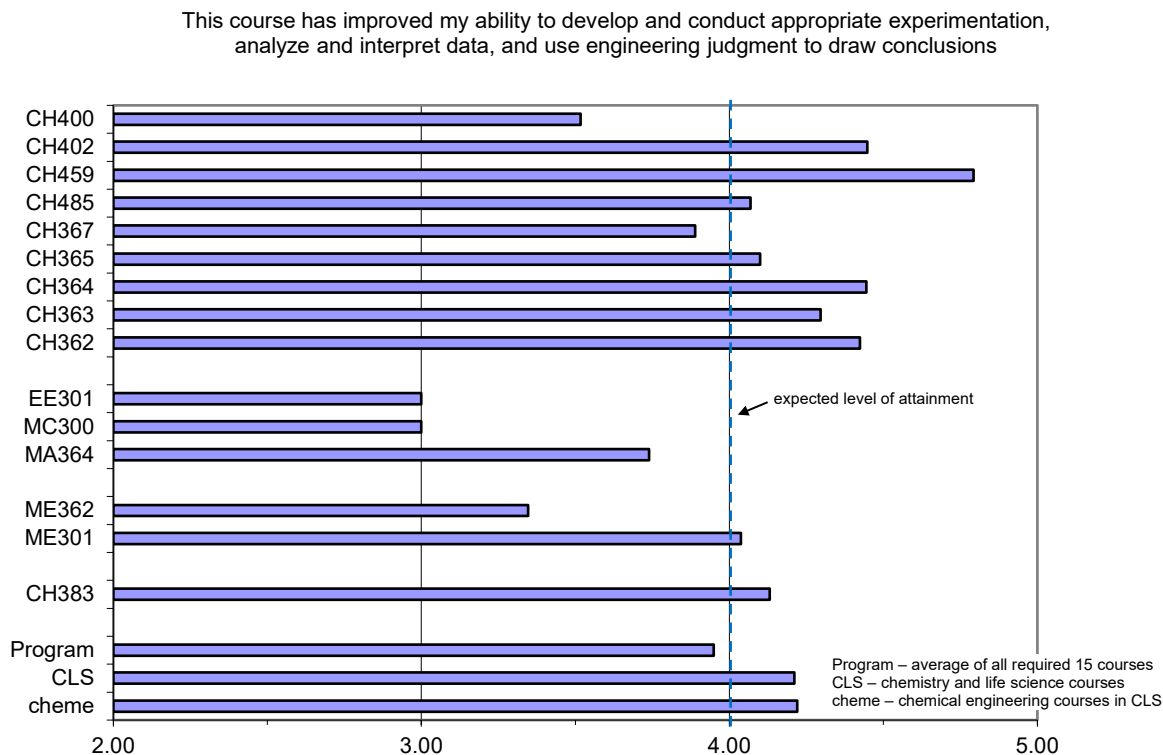


Figure 6-2. End-of-Semester Student Survey responses for Student Outcome 6.

3. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to the Firsties at the end of their last semester. They were asked whether or not they agree with the statement “The program has prepared me to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.” 29 out of 29 cadets completed the survey. Of these, 29 out of 29 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 21/29 replying that they strongly agreed (score = 5/5) and 8/29 that they agreed (score = 4/5). The mean is $4.72/5.00 \pm 0.46$ for the 29 cadets. The expected level of attainment on this survey is 4.00/5.00.
4. The average course grade in CH459 Chemical Engineering Laboratory was 3.52 ± 0.59 (n=29) in AY25, compared to 3.70 ± 0.46 (n=19) in AY24, 3.41 ± 0.43 (n=13) in AY23, 3.80 ± 0.42 (n=23) in AY22, 3.67 ± 0.53 (n=18) in AY21, and 3.53 ± 0.48 (n=29) in AY20. The 5-year running average is 3.62, and this is our expected level of attainment. This year’s score was *slightly below* the 5-year running average.

Level of Achievement of Student Outcome 7:

On completion of the chemical engineering program, our graduates will be able to acquire and apply new knowledge as needed, using appropriate learning strategies.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. Percent of eligible students taking the Fundamentals of Engineering Examination (FEE), once/yr.
3. End of Semester Student Surveys, once/semester.
4. Chemical Engineering Program Exit Survey, once/yr.
5. Lifelong Learning Skills Rubric, twice per year.
6. Contemporary Issues Rubric, multiple times per year.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

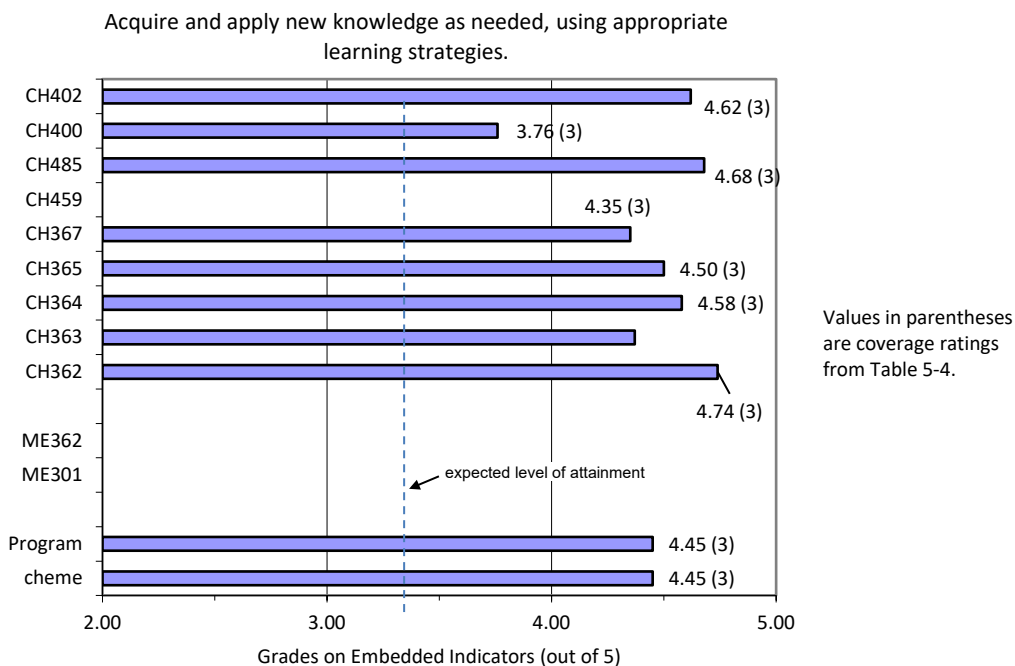


Figure 7-1. Coursework Embedded Indicator results for Student Outcome 7.

2. Percent of Eligible Cadets taking the Fundamentals of Engineering Examination (FEE). For the Class of 2025, 29 of 29 chemical engineering cadets (100% of eligible cadets) prepared for and took the FEE. For the Class of 2024, 19 of 19 chemical engineering cadets (100% of eligible cadets) prepared for and took the FEE and 100% in years 2021 to 2023. In 2020, 31% (9 of 29) took the exam. The decline in participation in 2020 was due to closure of NCEES test centers nationwide and to restrictions on cadet movement during the COVID-19 crisis. NCEES reported a decline from 1047 participants in 2019 to 480 in 2020, which recovered somewhat to 668 in 2021, 600 in 2022, 521 in 2023, 654 in 2024, and 674 in 2025. Our expected level of attainment is 100%.

3. End of Semester Student Surveys

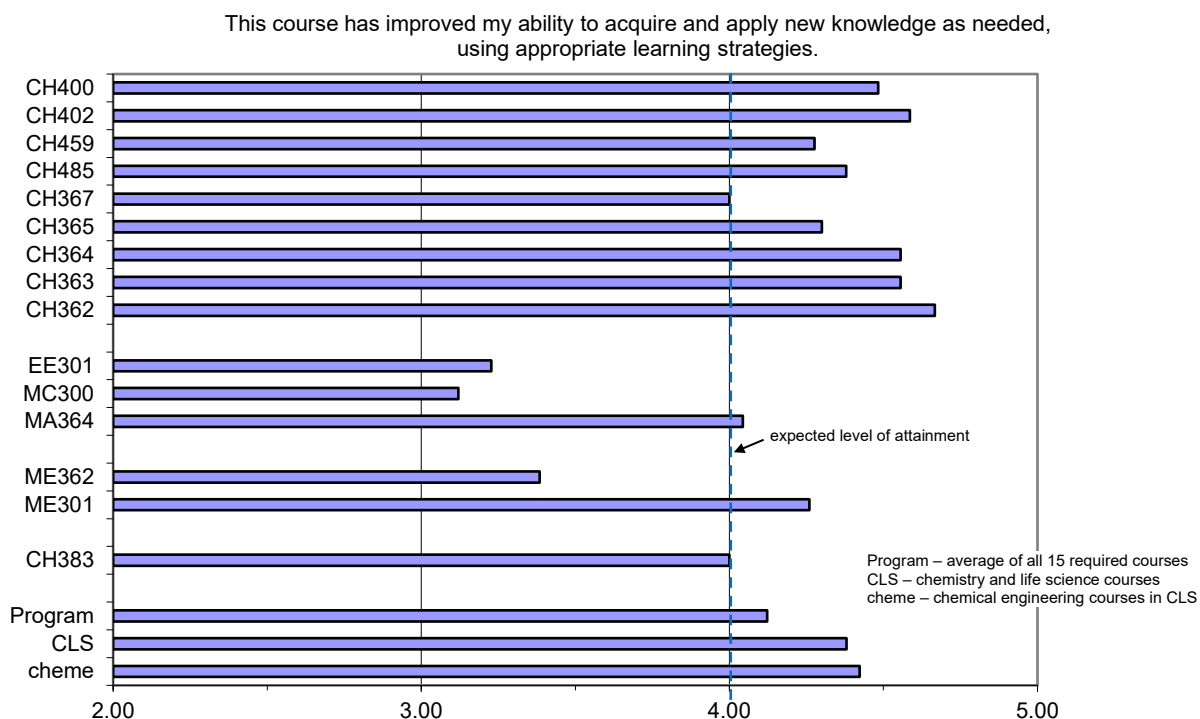


Figure 7-2. End-of-Semester Student Survey responses for Student Outcome 7.

4. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to the Firsties at the end of their last semester. They were asked whether they agree with the statement “The program has prepared me to acquire and apply new knowledge as needed, using appropriate learning strategies.” 29 out of 29 cadets completed the survey. Of these, 29 out of 29 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 22/29 replying that they strongly agreed (score = 5/5) and 7/29 that they agreed (score = 4/5). The mean is $4.76/5.00 \pm 0.44$ for the 29 cadets. The expected level of attainment on this survey is 4.00/5.00.
5. Lifelong Learning Skills Rubric. This rubric, when used in tandem with the resume writing assignment in CH365, is designed to assess performance in four skills associated with the ability to acquire new knowledge, namely: rubric row 1 - engagement (in professional activities), rubric row 2 - recognition (of skills learned in the program), rubric row 3 – intellectual growth (recognition of new skills), and rubric row 4 – communication (in resume format). Resumes were written at the beginning of the semester (assignment 1.1), then revised at the end of the semester (assignment 2.1). One rubric is shown for each assignment below, along with cadet averages and standard deviations. The expected levels of attainment are color-coded, with red indicating a need for improvement, yellow indicating acceptable performance, and green indicating that expectations are met or exceeded. Instructor comments are in the rubrics.

Instructor's Name: Biaglow				Cadet Assessed: Summary, all cadets in CH365				
Your Position: Instructor, CH365 (e.g., CD CH365)				Cadet Major: Chemical Engineering (e.g., Chem. Eng.)				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Engagement Outcome 7	No evidence of pre-professional activities.			References to pre-professional activities are lacking or connections to chemical engineering are weak or implied.			Uses examples of pre-professional chemical engineering activities.	4.8 ± 0.3
Recognition Outcome 7	Skills learned in chemical engineering courses taken in previous semesters are not listed.			Skills are listed, but the skills are vaguely described, or connection to chemical engineering concepts is not clear.		4.1 ± 0.7	Identifies specific skills learned in chemical engineering courses.	
Intellectual Growth Outcome 7	Unable to identify new concepts learned this semester.			Changes are apparent in document, but connections to recent activities in chemical engineering are weak or implied.	2.9 ± 1.2		Addition of multiple skills acquired this semester.	
Communication Outcome 3	Resume lacks organization or cohesion. Numerous grammatical errors that may interfere with meaning. Target audience unclear.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write a basic resume, but document is uninteresting and flat.			Demonstrates an ability to effectively communicate in the resume format. Clear, concise content. Resume is interesting.	4.4 ± 0.9
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) N	Comments: Cadet scores were somewhat low overall due to the low score in row three. Scores were low in this row because it is early in the semester and many cadets are unable to articulate new skills learned this semester.					Assignment used for assessment: (e.g., resume draft 1.1 in CH365) resume draft 1.1 in CH365		
	Final Grade: 4.1 ± 1.1 / 5.0							

Instructor's Name: Biaglow				Cadet Assessed: Summary, all cadets in CH365				
Your Position: Instructor, CH365 (e.g., CD CH365)				Cadet Major: Chemical Engineering (e.g., Chem. Eng.)				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Engagement Outcome 7	No evidence of pre-professional activities.			References to pre-professional activities are lacking or connections to chemical engineering are weak or implied.			Uses examples of pre-professional chemical engineering activities.	4.9 ± 0.3
Recognition Outcome 7	Skills learned in chemical engineering courses taken in previous semesters are not listed.			Skills are listed, but the skills are vaguely described, or connection to chemical engineering concepts is not clear.			Identifies specific skills learned in chemical engineering courses.	4.5 ± 0.5
Intellectual Growth Outcome 7	Unable to identify new concepts learned this semester.			Changes are apparent in document, but connections to recent activities in chemical engineering are weak or implied.		4.2 ± 1.0	Addition of multiple skills acquired this semester.	
Communication Outcome 3	Resume lacks organization or cohesion. Numerous grammatical errors that may interfere with meaning. Target audience unclear.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write a basic resume, but document is uninteresting and flat.			Demonstrates an ability to effectively communicate in the resume format. Clear, concise content. Resume is interesting.	4.8 ± 0.5
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) Y	Comments: Scores increased in all rows due to cadets having a chance to make revisions. the largest improvement is in row three. Cadets were better able to identify activities from this semester the demonstrate newly learned skills.					Assignment used for assessment: (e.g., resume draft 1.1 in CH365) resume draft 2.1 in CH365		
	Final Grade: 4.6 ± 0.7 / 5.0							

There is normally improvement in scores in draft 2.1 as cadets incorporate instructor comments to improve and polish their documents. Version 2.1, though, requires articulation of new skills acquired during the semester. Scores can go down if cadets do not show new activities and skills learned during the current semester, and in some individual cases the scores did drop, and intellectual growth (row 3) did increase. Overall and on average, the program scores are solidly in the green in all four categories by the end of the semester.

6. Contemporary Issues Rubric (follows on page 18). The contemporary issues rubric is designed to assess performance in four skills associated with the ability to acquire new knowledge, namely: rubric row 1 - acquiring knowledge (through development and use of references), rubric row 2 - technical competence of the cadet (as demonstrated with writing skills), rubric row 3 – impact of contemporary issues (ability to link contemporary issues to relevant chemical engineering concepts), and rubric row 4 - grammar and structure. Each of these skills, taken independently, are associated with written communication (Student Outcome 3). However, the development and blending of these skills into a coherent and well-crafted essay are a measure of acquisition of new knowledge and are thus used by our program to assess Student Outcome 8. Admittedly, this is not a complete measure of a student's ability to acquire new knowledge, only that ability as expressed in an essay. Therefore, we combine this exercise with the other embedded indicators in this section of the report.

Operationally, the contemporary issues rubric is completed by the instructor to award grades to cadets for the writing assignments in this course. In this case, LTC Cowart used one writing assignment in CH485, completed in two drafts. The draft and final submissions were graded against the rubric, with the total average score and average scores per outcome shown in the screenshots above. In AY25 (this year), the average for the draft was 4.1 ± 0.9 and the average on the final was 4.8 ± 0.6 , so cadets achieved a better average on the final than on the draft submission. Scores also showed some improvement year-to-year, compared to 4.0 ± 0.81 and 4.8 ± 0.4 in AY24 in the draft and final reports, respectively. We conclude that cadets achieved each of the graded rubric rows with proficiency. Instructor comments are included in the screenshots.

Instructor's Name: LTC Cowart				Cadet Assessed: All Cadets in CH485 (AY25-1)				
Your Position: CD, CH485 (e.g., CD CH485)				Cadet Major: Chemical Engineering (e.g., Chem. Eng.)				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Acquiring Knowledge Outcome 7	Cites references but they are not substantive or do not address assignment. Context is weak or implied. Uses generic arguments or lacks specificity.			Cites at least two references relevant to the assignment. References have been adequately studied and re-worded to support a thesis.			Uses multiple substantive examples or scholarly articles in an integrative fashion to support a thesis.	4.6 +/- 0.9
Technical Competence Outcomes 8	Demonstrates poor or incomplete understanding of technical content.			Demonstrates some knowledge of the technical content, but explanation lacks adequate depth.		3.6 +/- 0.9	Demonstrates exceptional knowledge of technical content.	
Impact of Contemporary Issues Outcome 4	Does not have a premise or does not connect issues in essay to concepts in chemical engineering.			Makes connections to chemical engineering concepts, but the connections are weak or implied or premise is weak.		3.9 +/- 0.8	Makes very clear connections between premise and chemical engineering concepts.	
Communication Outcome 3	Writing lacks organization or cohesion. Numerous grammatical errors or errors interfere with meaning. Thesis lacking or implied.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write an essay but lacks cohesion or completeness. Thesis not fully supported.		4.2 +/- 0.8	Demonstrates an ability to effectively communicate in the essay format. Fully supported, clear, concise thesis. Writing style was exceptionally clear and articulate.	
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) Y	Comments: Cadets met expectations on the draft writing assignment. Cadets demonstrated acquisition of new knowledge and generally wrote in an acceptable format. Technical content with respect to chemical engineering processes, including heat & mass transfer can be improved, as well as writing organization. Final Grade: 4.1 +/- 0.90						Assignment used for assessment: (e.g., draft 1 writing assignment 1 in CH485) CH485 Writing DRAFT	

Instructor's Name: LTC Cowart				Cadet Assessed: All Cadets in CH485 (AY25-1)				
Your Position: CD, CH485 (e.g., CD CH485)				Cadet Major: Chemical Engineering (e.g., Chem. Eng.)				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Acquiring Knowledge Outcome 7	Cites references but they are not substantive or do not address assignment. Context is weak or implied. Uses generic arguments or lacks specificity.			Cites at least two references relevant to the assignment. References have been adequately studied and re-worded to support a thesis.			Uses multiple substantive examples or scholarly articles in an integrative fashion to support a thesis.	4.8 +/- 0.4
Technical Competence Outcomes 8	Demonstrates poor or incomplete understanding of technical content.			Demonstrates some knowledge of the technical content, but explanation lacks adequate depth.			Demonstrates exceptional knowledge of technical content.	4.6 +/- 0.8
Impact of Contemporary Issues Outcome 4	Does not have a premise or does not connect issues in essay to concepts in chemical engineering.			Makes connections to chemical engineering concepts, but the connections are weak or implied or premise is weak.			Makes very clear connections between premise and chemical engineering concepts.	4.7 +/- 0.6
Communication Outcome 3	Writing lacks organization or cohesion. Numerous grammatical errors or errors interfere with meaning. Thesis lacking or implied.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write an essay but lacks cohesion or completeness. Thesis not fully supported.			Demonstrates an ability to effectively communicate in the essay format. Fully supported, clear, concise thesis. Writing style was exceptionally clear and articulate.	4.7 +/- 0.5
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) Y	Comments: Significant improvement from the draft version of the contemporary issues essay. Technical improvement, inclusion of figures/drawings/schematics, coupled with extensive discussion of contemporary issues regarding the chosen topic resulted in interesting, thoughtful papers. Final Grade: 4.8 +/- 0.6						Assignment used for assessment: (e.g., draft 1 writing assignment 1 in CH485) CH485 Writing FINAL	

Level of Achievement of Student Outcome 8:

The program provides the graduate with a thorough grounding and working knowledge of the chemical sciences, including:

- (8.1) Chemistry.
- (8.2) Material and energy balances
- (8.3) Safety and environmental factors.
- (8.4) Thermodynamics of physical and chemical equilibria.
- (8.5) Heat, mass, and momentum transfer.
- (8.6) Chemical reaction engineering.
- (8.7) Continuous and staged separation operations.
- (8.8) Process dynamics and control.
- (8.9) Modern experimental and computing techniques.
- (8.10) Process design.

Assessment Instruments and Frequency:

1. Fundamentals of Engineering Examination, once/yr.
2. Average Course Grades for Chemical Engineering Students, once/yr.

Assessment Results:

1. Fundamentals of Engineering Examination, Table 4-14. For the Class of 2025, the breakdown by topic is shown in comparison to the national averages.

Subject	Outcome	Questions	USMA ChE	National (expected level of attainment)
Chemistry & Biology	8.1	7	9.9	9.6 ± 2.7
Material & Energy Balances	8.2	10	9.1	9.4 ± 2.3
Safety, Health, & Environmental	8.3	5	10.1	9.9 ± 3.3
Thermodynamics	8.4	8	9.8	9.2 ± 2.2
Heat Transfer	8.5	8	9.6	9.7 ± 2.5
Fluid Mechanics/Dynamics	8.5	8	9.3	9.9 ± 2.7
Chemical Reaction Engineering	8.6	7	10.0	9.7 ± 2.6
Mass Transfer & Separations	8.7	8	9.9	9.5 ± 2.3
Process Control	8.8	4	8.6	9.8 ± 3.5
Economics	8.10	4	12.2	10.0 ± 3.7
Process Design	8.10	7	9.8	9.9 ± 2.7

2. Course grades for the last six years are shown below, Table 4-4 (from Self Study).

		Chemical Engineering Student Outcome 8								
Course ↓		Advanced Chemistry	Mater. & Energy Bal.	Thermodynamics	Transport	Reaction Engineering	Separations	Dynamics & Control	Experiment & Compute	Process Design
		8.1	8.2	8.4	8.5	8.6	8.7	8.8	8.9	8.10
CH383	Organic Chemistry I	2.97								
CH365	Chem. Eng. Thermo.			3.69						
CH362	Mass & Energy Balances		3.36							
CH363	Separation Processes						3.66			
CH364	Chem. Reaction Eng.					3.25				
CH459	Chem. Eng. Laboratory								3.52	
CH485	Heat and Mass Transfer				3.35					
CH400	Chemical Engineering Sem.									
CH402	Chem. Eng. Process Des.									3.34
MA366	Vector Calculus									
ME301	Thermodynamics			3.52						
ME362	Fluid Mechanics				3.55					
CE300	Fund. Eng. Mech. & Des.									
EE301	Intro. To Elec. Engineering									
CH367	Intr. Auto. Process Control							3.36		
Average Grade 2025		2.97	3.36	3.60	3.45	3.25	3.66	3.36	3.52	3.34
Average Grade 2024		2.53	3.48	3.64	3.23	3.54	3.68	3.28	3.70	3.65
Average Grade 2023		2.64	3.17	3.35	3.21	3.38	3.59	3.51	3.41	3.44
Average Grade 2022		3.15	3.64	3.59	3.47	3.33	3.69	3.42	3.80	3.33
Average Grade 2021		2.78	3.54	3.33	3.23	3.31	3.37	3.35	3.67	3.46
Average Grade 2020		3.06	3.53	3.27	3.20	3.38	3.64	3.36	3.53	3.26
Average Grade 2019		3.14	3.63	3.69	3.39	3.67	3.92	3.51	3.52	3.26
<i>Previous 5-year Running Average</i> (Expected level of attainment)		2.83	3.47	3.41	3.28	3.39	3.59	3.38	3.62	3.43
Standard Deviation 2025 (N=29)		1.04	0.60	0.84	0.67	0.60	0.63	0.70	0.60	0.70